

of connectivity or conductivity. How far a spanning tree can reach starting from any point in the graph provides information about the connectivity of the graph, while the number of possible spanning trees gives information about the number of possible paths, or the degree of connectivity.

For any graph of n nodes, Cayley's formula states that the maximum number of spanning trees in a bipartite graph (i.e. if the graph was a complete bipartite graph) is:

$$\tau_c(\mathbf{G}_S) = |M|^{(|C|-1)} |C|^{(|M|-1)} \quad (18)$$

with the weighted edge version, if we assume naively equal weights for all edges and still summing to one for every manuscript m being:

$$\tau_{cw}(\mathbf{G}_S) = s^{(|M|+|C|-1)} |M|^{(|C|-1)} |C|^{(|M|-1)} \quad (19)$$

The graph of shares $\mathbf{G}_S$ is expected to be very sparse, and therefore the actual number of tree spanning will be much lower. The actual number of weighted spanning trees, from Kirchhoff's matrix tree formula, is given by

$$\tau_{kw}(\mathbf{G}_S) = \det(\mathbf{L}_S^*) \quad (20)$$

where $\mathbf{L}_S^*$ is any cofactor of the Laplacian $\mathbf{L}_S$ of the graph $\mathbf{G}_S$, and can be formed by removing any one column and row from $\mathbf{L}_S$. Note that this version takes into the account the strength of connections between nodes. For an unweighted version of the number of spanning trees τ_k , set all non-zero values in $\mathbf{G}_S$ to be 1 before computing $\mathbf{L}_S^*$ and then τ_k .

Two spanning tree ratios can be then computed. The first is the unweighted ratio of the logarithm of spanning trees.

$$\text{STR} = \frac{\log(\tau_c)}{\log(\tau_k)} \quad (21)$$

This measure tells us how densely connected (or sparse) the shares graph is compared to a theoretical maximum where everyone collaborated on every paper with everyone else. It can be interpreted as a global collaborativeness coefficient, and can be tracked as a measure of academic trends and research health indicators. Note that the logarithm is used because we expect such extreme sparsity that the measure is only expected to have resolving power when logged.

The second ratio is the weighted ratio.

$$\text{STR}_w = \frac{\log(\tau_{cw})}{\log(\tau_{kw})} \quad (22)$$

This ratio tells us in relation to the first, the "conductivity" or the connectivity density modulated by the strength of connections rather than purely the presence of a connection. If this ratio is much lower than the first, it suggests many uneven share distributions rather than uniform like distributions. Thus the ratio of the two together can be thought of as a measure of asymmetry in share distribution or if taken as analogous to a conductive material, the purity of the conductor.

$$\text{RSTR} = \frac{\text{STR}_w}{\text{STR}} \quad (23)$$

3.8 Two-Step Shares Graph

The shares graph G_S shows how people and manuscripts are connected in one step (edge). To observe relationships between contributors, or between manuscripts, or if we want to separate the graph into two connectivity graphs (contributors, and manuscripts), a natural operation is raising G_S to the power of 2. This is called the two-step shares graph, with each element showing the equivalent of random walk likelihood or density of connections from manuscript to manuscript, or contributor to contributor.

$$\mathbf{G}_s^2 = \mathbf{G}_S^T \mathbf{G}_S = \mathbf{G}_S \mathbf{G}_S \quad (24)$$

The adjacency matrix of the two step shares graph has a block-diagonal structure, with two major blocks on the diagonal corresponding to the two separate subgraphs:

$$\mathbf{G}_s^2 = \left[\begin{array}{c|c} \mathbf{M}_S^2 & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{C}_S^2 \end{array} \right]$$

- $\mathbf{M}_S^2$ which has elements representing likelihood of random walking from a manuscript node to a manuscript node along the edges on G_S .

The diagonal elements of this component $\mathbf{M}_S^2(x, x) = \mathbf{M}_S(x, :) \cdot \mathbf{M}_S(x, :)$ represent how concentrated (non-uniform) the shares distribution is for the manuscript m_x for its contributors.

The off-diagonal elements $\mathbf{M}_S^2(x, y) = \mathbf{M}_S(x, :) \cdot \mathbf{M}_S(y, :)$, represent the similarity of contributor shares between manuscripts m_x and m_y , with a maximum value (1) attained by an exact match in share split, and a minimum value of (0) attained by no shared contributors.

- $\mathbf{C}_S^2$ which has elements representing likelihood of random walking from a contributor node to a contributor node along the edges on G_S . The $\mathbf{C}_S^2$ is composed of 9 sub-blocks, each with similar but distinct meanings for their values. The interpretations of these block element values are as follows.

- $\mathbf{S}_A^2$ represents traversal likelihood from an author node to an author node.

The diagonal elements of the block, gives the concentration of distribution of shares by the author across their manuscripts.

$$\mathbf{S}_A^2(x, x) = \mathbf{S}_A(x, :) \cdot \mathbf{S}_A(x, :), \quad \forall x \in \{x \in \mathbb{N}, |M| < x \leq |M| + |C|\}$$

The max value here A higher value here means the author's average shares are

The off-diagonal elements of the block, $\mathbf{S}_A^2(x, y) = \mathbf{S}_A(x, :) \cdot \mathbf{S}_A(y, :)$, $\forall |M| < x, y \leq |M| + |C|$, represent similarity of shares split between collaborating authors c_x and c_y across all manuscripts

- $\mathbf{S}_A \mathbf{S}_P$ represents traversing from an author node back to a peer reviewer node.

The diagonal elements of the block, $\mathbf{S}_A \mathbf{S}_P(x, y) = 0$, $\forall |M| < x \leq |M| + |C|, y = |M| + |C| + x$, since this would mean the same contributor was both an author and a peer reviewer for a manuscript.

The off-diagonal elements of the block, $\mathbf{S}_A \mathbf{S}_P(x, y) = \mathbf{S}_A(x, :) \cdot \mathbf{S}_P(y, :)$, $\forall |M| < x \leq |M| + |C|, |M| + |C| < y \leq |M| + 2|C|$, represent similarity of shares split between c_x and c_y when c_x is the author and c_y is the peer reviewer, across all manuscripts where that holds true.

- $\mathbf{S}_A \mathbf{S}_R$ represents traversing from an author node back to a replicator node. The diagonal and off-diagonal elements hold the same interpretations as for the $\mathbf{S}_A \mathbf{S}_P$ block

$$\mathbf{G}_S^2 = \begin{bmatrix} \mathbf{M}_S^2 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_A^2 & \mathbf{S}_A \mathbf{S}_P & \mathbf{S}_A \mathbf{S}_R \\ \mathbf{0} & \mathbf{S}_A \mathbf{S}_P & \mathbf{S}_P^2 & \mathbf{S}_P \mathbf{S}_R \\ \mathbf{0} & \mathbf{S}_A \mathbf{S}_R & \mathbf{S}_P \mathbf{S}_R & \mathbf{S}_R^2 \end{bmatrix}$$

The concentration and similarities of shares distributions computed for contributors in different roles and manuscripts would serve as building blocks for metrics elaborated in section 8. Further, spectral analysis on the $\mathbf{G}_S^2$ graph, would help in looking at the spectral properties of connectivities between contributors only, and manuscripts only.

3.9 Condensed Form

The full adjacency matrix $\mathbf{G}_S$ contains a significant amount of structural redundancy due to its bipartite construction. In particular, 10 of the 16 blocks are identically $\mathbf{0}$, and the remaining nonzero blocks consist of three contributor–manuscript matrices and their transposes.

A condensed representation of the shares graph can therefore be defined by retaining only the 3 top-right blocks (out of 16) of $\mathbf{G}_S$:

$$\mathbf{G}'_S = [\mathbf{S}_A \quad \mathbf{S}_P \quad \mathbf{S}_R] \quad (25)$$

where:

- $\mathbf{S}_A$ is the matrix block that encodes shares held by author nodes,
- $\mathbf{S}_P$ is the matrix block that encodes shares held by peer reviewer nodes,
- $\mathbf{S}_R$ is the matrix block that encodes shares held by replicator nodes.

This condensed form captures all edge information in G_S , since the remaining nonzero blocks of $\mathbf{G}_S$ are given by the transposes $\mathbf{S}_A^T$, $\mathbf{S}_P^T$, and $\mathbf{S}_R^T$. The full adjacency matrix can therefore be reconstructed as:

$$\mathbf{G}_S = \begin{bmatrix} \mathbf{0} & \mathbf{G}'_S \\ (\mathbf{G}'_S)^T & \mathbf{0} \end{bmatrix} \quad (26)$$

The advantage of this representation is computational. Storing only $\mathbf{G}'_S$ reduces memory usage and simplifies matrix operations, while preserving the ability to recover the full graph when required for spectral or other analyses introduced in the above subsections.

4 References Graph

The references graph G_W is composed of manuscripts M as nodes, and citations as edges. The edges are denoted by U if they are unweighted, and W if they are weighted. In the traditional unweighted case, there would be a directed edge u with weight 1 going from manuscript node m_y to m_x if m_y cites m_x , while in the weighted case, the edge will have a weight $w_{x,y} \in \mathbb{R}$. Formally, $G_W = (M, W)$. This would normally yield a directed acyclic graph (Directed Acyclic Graph) (assuming no revisions are made). In Liberata, all revisions will be version controlled and citable.

As mentioned in section 2.2.2, in the Liberata system, each paper could only give out a single unit of citation or academic capital ($\mathbf{x}$). This upperbounds the total citation units disbursed by a manuscript to 1, while in the unweighted case, the total citation units disbursed by a manuscript is equal to the count of all the references r . More advanced methods of weighting have been explored in 10.

4.1 Matrix Representation

The references graph G_R can also be represented as a square adjacency matrix $\mathbf{G}_R \in \mathbb{R}^{|M| \times |M|}$, and each element $\mathbf{G}_W(x, y)$ will be given by

$$\mathbf{G}_W(x, y) = \begin{cases} w_{x,y} & \text{if } m_y \text{ cites } m_x \\ 0 & \text{otherwise} \end{cases} \quad (27)$$

Here, m_x, m_y are manuscripts and $w_{x,y}$ is the weighted citation received by m_x from its reference by m_y . This matrix will be an upper triangular matrix in the Frobenius normal form as shown below. In the simplest case of weighting with the inverse of citations as given in eq. (6), the adjacency matrix becomes column-stochastic as each column sums to 1.

$$\mathbf{G}_W = \begin{bmatrix} 0 & w_{1,2} & \cdots & w_{1,y} & \cdots & w_{1,n-1} & w_{1,n} \\ 0 & 0 & \cdots & w_{2,y} & \cdots & w_{2,n-1} & w_{2,n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & w_{x,y} & \cdots & w_{x,n-1} & w_{i,n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & w_{n-1,n} \\ 0 & 0 & \cdots & 0 & \cdots & 0 & 0 \end{bmatrix}$$

The Frobenius normal form would induce a temporal ordering along the rows and columns of the matrix. In this form, the DAG would be an upper triangular matrix with zero along the diagonals, since a manuscript cannot cite itself, and each manuscript may only be cited by other manuscripts published later than itself. Interesting to note, a lot of the bibliometry-based citation weights, e.g. co-citations or co-references could still be computed from two-step graphs (refer section 4.8 and section 4.9) when using the matrix representation.

4.2 Fetch Vectors

The value of the edge $w_{x,y}$ induces interesting properties when unit vectors $\mathbf{v}$ are multiplied onto the $\mathbf{G}_W$. Since each column contains the relative references of a single manuscript, the column sum will always be 1. $(\mathbf{G}_W \mathbf{v}_x) \cdot \mathbf{1}^{|M|} = \sum_x w_{x,y} = 1$. This serves as a quick integrity check of the references graph in case there are faulty or misspelled references.

A left multiplication with the fetch vector $(\mathbf{v}_y \mathbf{G}_W)$ results in a vector with components representing academic capital earned by y from every other work, $\mathbf{x}_y$. When weighted using eq. (6), these components represent a distribution of academic capital earned by manuscript y across the works that cite m_y and the component sum yields the total academic capital value accrued by the manuscript m_y , $\mathbf{x}_y$ as shown in eq. (28).

$$\mathbf{x}_y = \sum (\mathbf{v}_y \mathbf{G}_W) \quad (28)$$

Furthermore, since each fetch vector now yields a continuous set of numbers, it would be possible to make observations on more fine-grained assessments of the impactfulness of a manuscript by examining the distribution of impact of that manuscript on other manuscripts.